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Four weeks of sprint interval training improves 5 km run performance

Short title: SIT improves 5 km run performance

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ABSTRACT

Sprint interval training (SIT) rapidly improves cardio-respiratory fitness but demands less training time and volume than traditional endurance training. While the health and fitness benefits caused by SIT have received considerable research focus, the effect of short-term SIT on 5 km run performance is unknown. Thirty healthy untrained participants (aged 18–25 years) were allocated to a control ($n = 10$) or a SIT ($n = 20$) group. SIT involved three to eight sprints at maximal intensity, three times a week for four weeks. Sprints were progressed to eight by the 12th session. All participants completed a 5 km time-trial on a public running track and an incremental treadmill test in an exercise physiology laboratory to determine 5 km run performance and maximum oxygen uptake, respectively, before and after the four week intervention. Relative to the controls, sprint interval trained participants improved 5 km run performance by 4.5% ($p < 0.001$) and this was accompanied by improvements in absolute and relative maximum oxygen uptake (4.9%, $p = 0.04$ and 4.5%, $p = 0.045$, respectively). Therefore, Short-term SIT significantly improves 5 km run performance in untrained young men. We believe SIT is a time-efficient means of improving cardio-respiratory fitness and 5 km endurance performance.

KEYWORDS: Time-trial, training load, SIT, VO_{2max}

42 INTRODUCTION

43 Endurance run performance is important for individuals of average cardio-respiratory fitness
44 who aspire to improve their fitness and performance quickly and efficiently. Moreover,
45 considering the importance of cardio-respiratory fitness and endurance performance in ball
46 sports such as soccer, Australian Rules Football and cricket (5, 15, 18) there is immense
47 coaching emphasis on reducing training load to minimise injury risk, but maximise training
48 time to focus on other skills important to the sport. SIT is an alternative mode of training to
49 traditional constant-rate training as SIT improves cardio-respiratory fitness to a similar
50 degree as constant-rate training but requires 90% less weekly energy expenditure and 66%
51 less total training time (7).

52 Typically SIT consists of short, maximal efforts of 15–30 s, repeated several times
53 interspersed with 3–5 mins of recovery. Indeed, SIT may be more effective at improving
54 endurance performance than both continuous running and other high-intensity interval
55 training (HIT) programmes. For example, 20 individuals who completed seven to twelve
56 efforts of 30 s sprints thrice weekly for six weeks, improved 3 km run performance greater
57 than individuals who completed constant rate running or four to six four minute efforts at 3
58 km average running velocity, over six weeks (4). Others have, however, demonstrated
59 recreationally active participants similarly improve 2 km run performance after either six
60 weeks of SIT (4.6%) or constant rate running (5.9%) (13). Therefore, SIT seems to enhance
61 run performance over 2-3 km after six weeks of training, but whether these endurance
62 benefits improve run performance in untrained participants in shorter time-frames and over
63 distances exceeding 3 km (a distance requiring greater reliance on aerobic metabolism) is
64 unknown. Furthermore, the lack of inclusion of experimental controls in previous studies (4,
65 6, 13) obfuscates the genuine impact of SIT on endurance run performance.

66

67 Therefore, the purpose of our study was to determine whether a novel and relatively short-
68 term (four week), thrice weekly SIT programme improves cardio-respiratory fitness and 5 km
69 run performance. To that end, we assessed 5 km run performance and $\dot{V}O_{2\max}$ before and
70 after either a novel four week SIT (four to eight 30-s maximal sprints, thrice weekly)
71 programme for cases (n = 20), or no training for controls (n = 10). We hypothesised that
72 relative to the controls, the sprint-interval trained individuals would significantly improve 5
73 km run performance and $\dot{V}O_{2\max}$.

74

75 METHODS

76 *Experimental approach to the problem*

77 This study is a controlled trial, where healthy participants performed either four weeks of SIT
78 or no training, to evaluate the effect of SIT on 5 km run performance and $\dot{V}O_{2\max}$. $\dot{V}O_{2\max}$
79 was measured during a maximal treadmill test conducted in the university exercise
80 physiology laboratory and the 5 km time-trial was assessed at a public running track. These
81 tests were performed on separate days within one week of commencing the initial SIT session
82 and within one week after the final SIT session.

83

84 *Subjects*

85 Thirty, apparently healthy young men (18–25 years) were recruited for this study.
86 Participants were initially screened to ensure they were not currently engaging in any
87 structured high-intensity aerobic exercise training. All participants had not completed any

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88 structured aerobic exercise training in the past year. Twenty participants were allocated to the
89 SIT (cases) group and 10 participants were allocated to the control group.

90

91 Participants gave written informed consent and this study was approved by the University's
92 Human Research Ethics Committee.

93

94 *Procedures*

95 Participant $\dot{V}O_{2\max}$ was assessed during a maximal treadmill test, by pulmonary analysis
96 conducted in an exercise physiology laboratory at the university. Before the $\dot{V}O_{2\max}$ test
97 participants were fitted with a two-way breathing valve (Hans Rudolph) and expired air was
98 collected into an online metabolic system (Moxus) for gas (O_2 and CO_2) analysis. The
99 metabolic system was calibrated before each test using ambient air and gas of known
100 composition. Participants were given a standardised 3-min warm-up at $10\text{ km}\cdot\text{h}^{-1}$. The $\dot{V}O_{2\max}$
101 test commenced at $10\text{ km}\cdot\text{h}^{-1}$, and treadmill speed was progressively increased by $1\text{ km}\cdot\text{h}^{-1}$
102 every second minute until volitional exhaustion. $\dot{V}O_{2\max}$ was determined as the highest O_2
103 value averaged over 60-s. [ENREF 7](#) All laboratory testing was performed preprandially at the
104 same time of day (8–10 AM). Participants were encouraged to hydrate the night before and
105 morning of testing. Within a week of the $\dot{V}O_{2\max}$ assessment each participant completed a
106 supervised 5 km run time-trial on a flat running track in a local park. Participants were
107 familiar with 5 km circuit and were advised to hydrate before running. All testing was
108 conducted in the afternoon (4–6 PM). Briefly, participants completed a short 10-min warm-
109 up including some light aerobic exercise and dynamic stretches. Participants were supervised
110 and instructed to run maximally at their own pace.

111

112 Cases completed a standardised SIT programme performed three times a week over four
113 weeks (total of 12 sessions). The sprint duration and recovery period was controlled at 30-s
114 and 4-mins (passive), respectively. Participants were requested to run maximally without
115 pacing for each 30-s sprint. All training was conducted on the University's sports oval and an
116 Accredited Exercise Physiologist (with Exercise and Sport Science Australia) provided
117 participants with verbal encouragement, and supervised each training session. The SIT
118 volume increased from four to eight sprints over the four week intervention (Table 1). Before
119 each training session, participants completed a standardised warm-up entailing a 5-min
120 aerobic warm-up, dynamic stretches and some short (20 m) runs at approximately 70, 80, 90
121 and 100% effort.

122

<<Table 1 about here>>

123 All participants were instructed not to deviate from their current physical activity, exercise (if
124 any) and dietary habits during the four week intervention period.

125 *Statistical analyses*

126 All statistical analyses were performed using IBM SPSS Statistics for Windows (Version 20,
127 IBM Corp, Armonk, NY). Data were tested for normality using the Kolmogorov-Smirnov
128 and Shapiro-Wilk tests, and non-parametric data was log-transformed before further analysis.
129 Paired samples *t*-tests were used to examine fitness and performance changes after SIT,
130 relative to the control group. To ascertain the test-retest reliability of $\text{VO}_{2\text{max}}$ and 5 km time-
131 trial performance, the intraclass correlation coefficient (2, k), coefficient of variation (log-
132 transformed), standard error of measurement, and systematic bias (determined by paired
133 samples *t*-test) were calculated. Significance was set at $p < 0.05$.

RESULTS

Test-retest reliability data is outlined in Table 2. The 5 km time-trial test had excellent reliability with an intraclass correlation coefficient of 0.99 and a 3.4% coefficient of variation.

<<Table 2 about here>>

Figure 1 outlines the fitness and performance changes after four weeks of SIT. Relative to the control group who showed a marginal increase in 5 km time-trial performance (mean $\pm$ SD: 1478 $\pm$ 350 to 1447 $\pm$ 347, -2%, $p = 0.20$, Table 2 and Figure 1), cases had a significant improvement in 5 km time-trial performance by an average of 65 s (mean $\pm$ SD: 1464 $\pm$ 298 to 1368 $\pm$ 270, -4.4%, $p < 0.001$) following four weeks of SIT (Figure 1). While absolute and relative $\dot{V}O_{2\max}$ increased after SIT in cases (mean $\pm$ SD: 3.8 $\pm$ 0.6 to 4.0 $\pm$ 0.6, 4.9%, $p = 0.04$ and 49.6 $\pm$ 4.6 to 51.4 $\pm$ 3.8, 4.5%, $p = 0.045$, respectively), absolute and relative $\dot{V}O_{2\max}$ were unchanged in controls (mean $\pm$ SD: 3.7 $\pm$ 0.6 to 3.7 $\pm$ 0.6, 0.6%, $p = 0.73$ and 49.6 $\pm$ 6.7 to 49.3 $\pm$ 6.7 -0.6%, $p = 0.72$, respectively, Table 2 and Figure 1).

<<Figure 1 about here>>

DISCUSSION

The purpose of our study was to determine whether a novel and short-term (four week), thrice weekly SIT programme improves cardio-respiratory fitness and 5 km run performance in untrained males. To our knowledge, we are the first to demonstrate that 5 km run performance is significantly improved following short-term (four weeks), low-volume SIT. Considering the marginal but not statistically significant improvement in the controls (31 s, 2%), the marked (96 s, 6.5%) improvement after four weeks of SIT translated to a mean 65 s

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(4.4%) faster 5 km run performance in cases. Notably, 5 km run performance was improved in conjunction with increased $\dot{V}O_{2\max}$ (4.5%).

Our data showing SIT enhances 5 km run performance corroborates others' showing SIT enhances 2 km (13) and 3 km (4, 6) run performance in recreationally trained individuals. We are the first to demonstrate four weeks of three times per week SIT, in the form of running, significantly improves 5 km run performance – a prestigious and Olympic running distance. Interestingly, the impact of SIT on 5 km endurance performance is established in other exercise modalities. A short-term (two-week) cycle SIT intervention improved 5 km cycle performance to a similar magnitude to our protocol (5.2%) (10). Whether SIT improves run performance over longer distances warrants attention. Additionally, whether SIT benefits already well-trained runners is yet to be fully understood, as data has, to date, been equivocal (1, 11). Nevertheless, we verify SIT as an effective means of improving running endurance performance in untrained young men.

The increased 5 km run performance in our study is similar to previously observed improvements after traditional constant-rate endurance training. It was reported, an intervention of three running sessions per week at 75% of $\dot{V}O_{2\max}$ for six weeks improved 5 km run time by approximately 80 s (~5%) in 39 untrained individuals (20). Additionally, others have shown 5 km time-trial improved by 78 s (5%) in a group that ran for 20 mins three times per week for six weeks, initially starting at 0.8 km·hr⁻¹ below their individual lactate threshold speed with progression to 0.8 km·hr⁻¹ above their individual pre-training lactate threshold speed (14). Importantly, a control group was not included in these studies to establish the error of the 5 km time-trial test, which we established was 31 s. Collectively, it

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appears SIT improves 5 km run performance to a similar extent as constant-rate training, but in a quicker time frame (four vs. six weeks) and a smaller training duration (249 mins vs. 360 mins) (14).

[ENREF 3](#)

The improvement in run performance was accompanied by a corresponding improvement in $\dot{V}O_{2\max}$. Although previous constant-rate training studies have demonstrated participants increased their $\dot{V}O_{2\max}$ to a similar magnitude observed in the our study (3.8–7.7%), they also showed marked improvements in their participants lactate threshold, which could have contributed to the improvements in run performance (14, 20), particularly because not all cases had improvements to $\dot{V}O_{2\max}$. Alternatively, an improvement in cardiac output may have facilitated the improvement in 5 km run speed, as opposed to improvements in muscle oxidative capability, based on the concepts $\dot{V}O_{2\max}$ is chiefly governed by the maximal cardiac output and the lactate threshold by muscle oxidative capability (12). Other mechanisms by which SIT may enhance 5 km run performance include improved run economy, running mechanics and potassium regulation. It was previously reported that moderately-trained runners' run economy was improved by 7% after four weeks of SIT, but these subject did not improve their 10 km time-trial performance (11). Moreover, increases in knee flexor endurance, coupled with decreases in knee flexion torque and knee flexion/extension ratios are also adaptations associated with sprint training that could have contributed to the improvement in 5 km run performance observed in our study (19). Others have shown in well-trained runners, SIT enhanced key components of the Na^+/K^+ pump that would aid to minimise disturbances in nerve membrane potential and maintain sprint performance, and this was associated with an approximate 3% improvement in 3 km and 10km run performance (1). The increase in muscle oxidative and anaerobic enzyme activity

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are adaptations gained from SIT and is another possible mechanism for the facilitated endurance performance observed in participants from our study (2, 3, 8, 16).

A limitation of our study is that we did not include females. Some of the benefits gained from SIT seem to be dependent of gender. For example, males had greater muscle protein synthesis of proteins important for mitochondrial biogenesis after nine sessions of SIT compared to their female counterparts (17). Given these muscle adaptations are vital for endurance performance (2, 8, 9), whether short-term SIT improves 5 km run performance in females is left for future investigations.

Future research could focus on optimising training variables, such as the recovery time between sprints, number of sprint repetitions, time required of each sprint, number of sessions required per week and the optimal length of SIT to improve fitness and performance. In fact, the efficiency of HIT in evoking positive adaptations may even be more potent than originally envisaged. It was recently revealed one maximal 4-min effort at 90% of heart rate maximum on a cycle ergometer, performed three times a week for 10 weeks improves $\dot{V}O_{2\max}$ by an average of 10% (21).

In conclusion our data reveals SIT is a highly effective strategy to improve endurance running quickly in previously untrained young men.

226 PRACTICAL APPLICATIONS

227 Physical conditioning professionals must be aware of efficient means of exercise training that
228 augment cardio-respiratory fitness and endurance performance. The results from our study
229 support the use of short-term, low-volume SIT as an effective means for rapidly improving
230 cardio-respiratory fitness and endurance performance. Therefore, team sport coaches should
231 consider incorporating short-term SIT within periodised programmes to improve athlete
232 endurance performance quickly and also to prevent musculoskeletal overuse injuries – a risk
233 associated with traditional forms of exercise that rely on large training volumes to improve
234 endurance performance. What is more, SIT has an added advantage of simultaneously
235 enhancing anaerobic qualities required in team sports. Finally, SIT should be encouraged to
236 individuals who perceive time as an obstacle to exercising, as four weeks of thrice weekly
237 SIT consisting of four to eight 30-s sprints (just over 4 hrs of total exercise, including rest
238 periods) quickly enhances cardio-respiratory fitness and endurance performance.

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FIGURE LEGENDS

Figure 1. 5 km time-trial performance and relative $\dot{V}O_{2\max}$ before and after SIT.

Relative to the controls (-2.1, $p = 0.20$), the cases had a significant improvement to 5 km run performance (-4.4%, $p < 0.001$) and relative $\dot{V}O_{2\max}$ (4.5%, $p < 0.05$) after SIT. Data are expressed as mean values before and after either four weeks of SIT (cases) or no exercise training (controls).

Legend: * $p < 0.05$; *** $p < 0.001$.

320 **TABLES**321 **Table 1.** Description of the four-week SIT programme.

Week	Session #	Training load (sprints)	Training sprint time (min)	Total session time (min)
1	1	3	1.5	9.5
	2	4	2	14
	3	5	2.5	18.5
2	4	4	2	14
	5	5	2.5	18.5
	6	6	3	23
3	7	5	2.5	18.5
	8	6	3	23
	9	7	3.5	27.5
4	10	6	3	23
	11	7	3.5	27.5
	12	8	4	32
Total				
time		66	33	249

322 Legend: # number.

323 **Table 2.** Test-retest reliability data for the 5 km time-trial test.

Variable	Mean trial 1	Mean trial 2	Mean difference	<i>p</i> -value	SEM	ICC	CV (%)
VO _{2max} (ml·kg ⁻¹ ·min ⁻¹)	49.57 ± 6.71	49.29 ± 6.67	-0.28	0.72	1.69	0.967 (0.867–0.992)	3.6 (2.5–6.7)
5 km time-trial (s)	1478.10 ± 349.95	1447.50 ± 347.24	-30.6	0.18	47.07	0.991 (0.963–0.998)	3.4 (2.4–6.4)

324

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325 Results of the 5 km time-trial test in the control group ($n = 10$),
326 reliability as measured by mean difference of two trials,
327 intraclass correlation (ICC), p -value (two-tailed paired t -test),
328 standard error of measurement (SEM) and typical error as a
329 coefficient of variation (CV) percentage (%). Upper and lower
330 confidence intervals are expressed in parentheses; confidence
331 intervals are set at 95%. SDs are represented after the mean of
332 trials 1 and 2

